Simulation for beginners - Project 1
Mobile drug testing at techno parties

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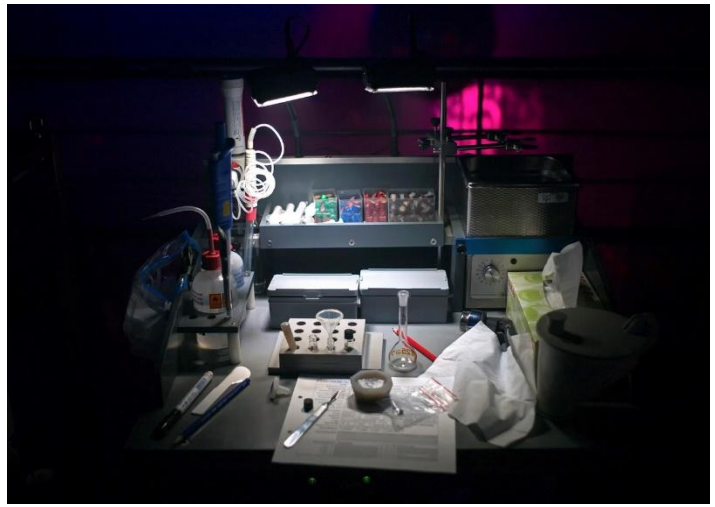
Key figures mobile drug testing

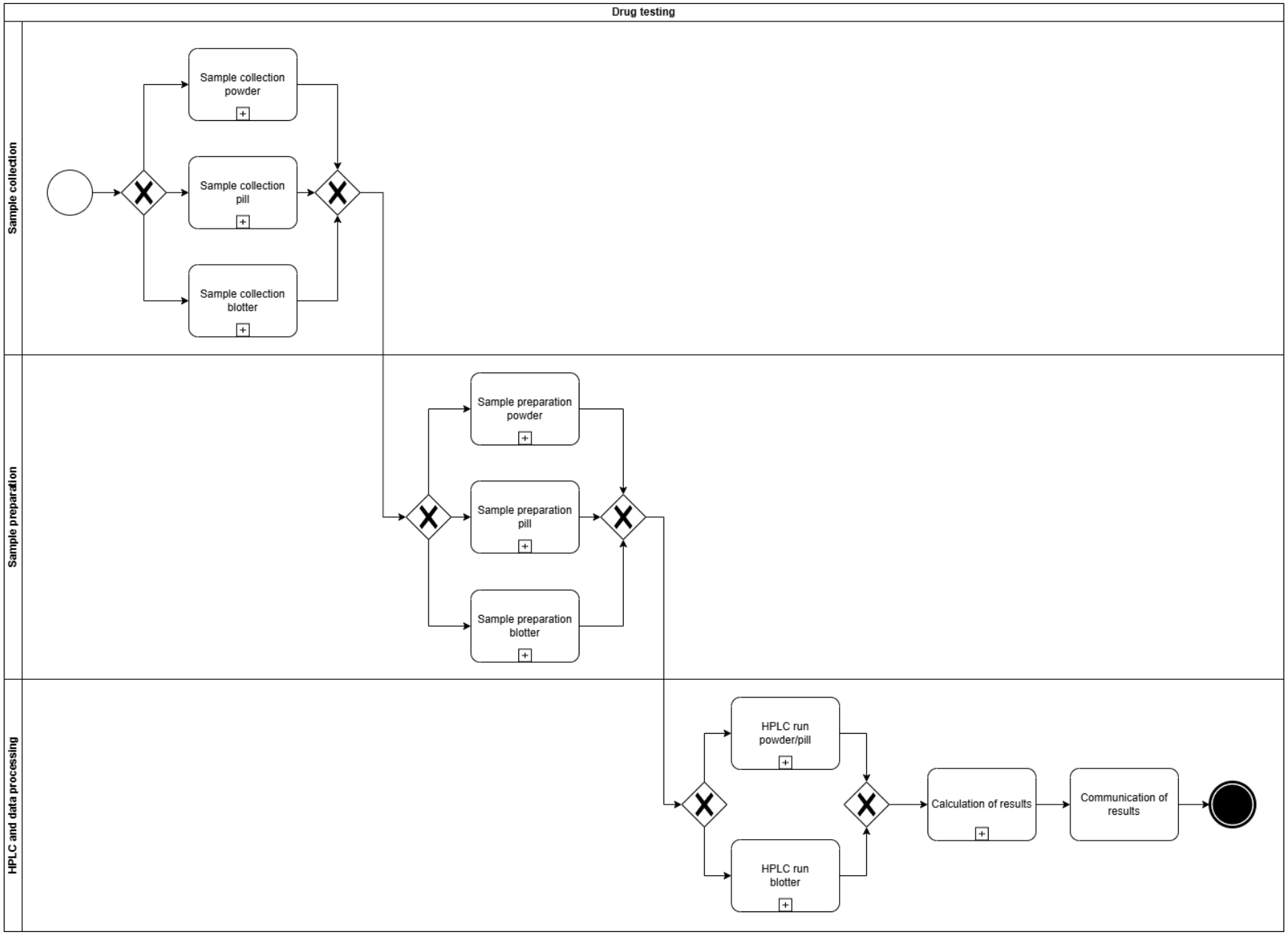
- Since 1998
- 6000 samples
- 250 events
- 160 chemical substances
- 5 harm reduction organisations
- 60 % of the samples result in warnings to clients

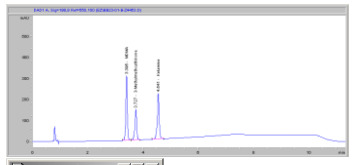
The equipment before loading into the vehicle

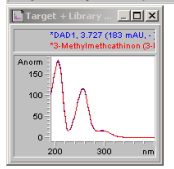
Mobile lab and counselling desk on the dance floor



BPMN diagram with pool / lanes







Symbol	Last	Change	%Chg	High	Low
BTC	32197.40	+1927.40	+6.4%	32197.4	30492.83
ETH	2161.73	+61.73	+2.9%	2161.73	2111.39
USDT	0.9944	-0.0056	-0.6%	1.0004	0.9944
BNB	345.53	+13.53	+4.1%	345.53	332.57
USDC	0.9959	+0.0009	+0.1%	0.9959	0.9948
XRP	0.5282	+0.0082	+1.6%	0.5282	0.5218
ADA	0.4475	-0.0049	-1.1%	0.4519	0.4455
DOGE	0.0886	0	0%	0.0887	0.0885
MATIC	23.96	-0.17	-0.7%	24.14	23.96
SOL	33198.98	+1198.98	+3.7%	33276.79	32383.09
DOT	6.89	+0.07	+1.0%	6.89	6.82
LTC	99.88				

KPI / Objective

- Number of samples per event
- Waiting time of the clients
- Staff utilisation rate
- Costs (HPLC devices / staff)

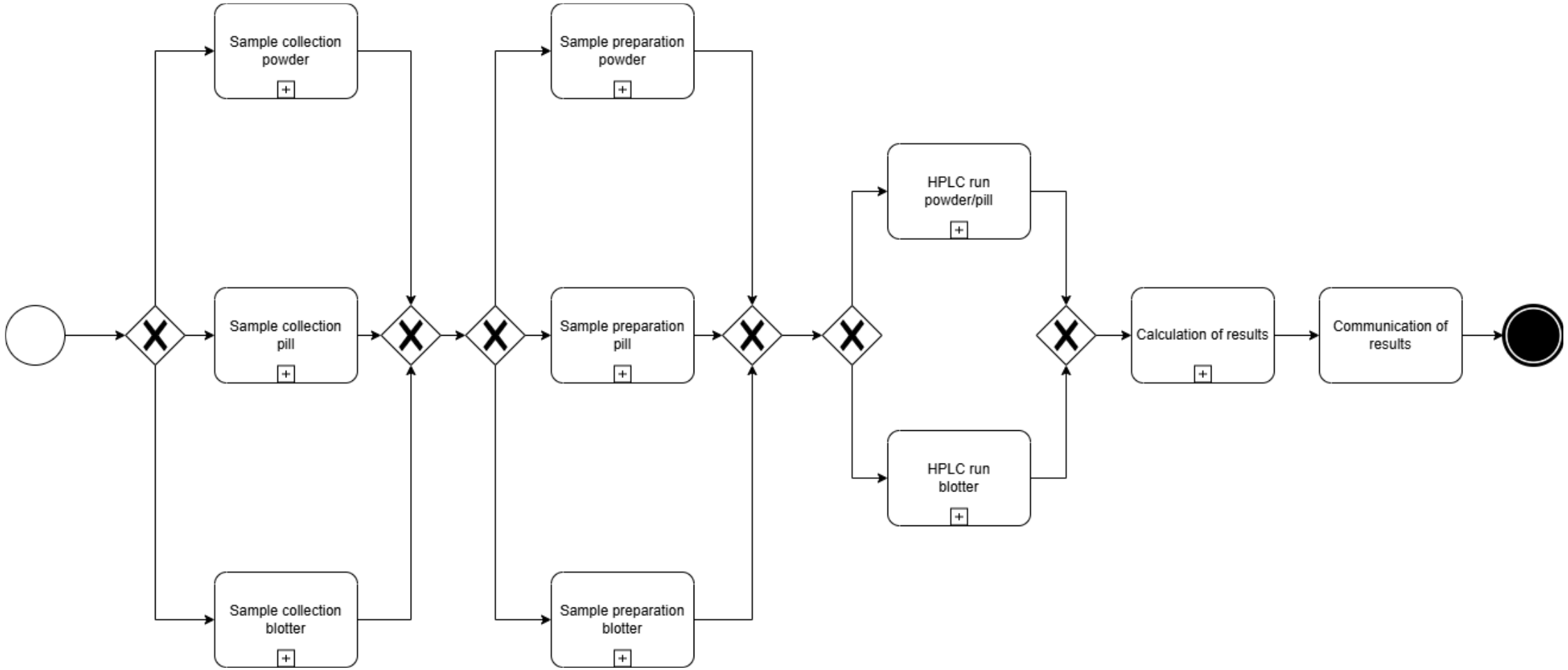
Parameters

- Sample arrival rate
- Processing time at each station
- Number of employees (per role)
- Number of HPLC devices
- Measuring time HPLC methods
- Distribution / percentage of sample types (powders / pills / blotters)
  → processing time sample collection and preparation
  → HPLC method → measuring time HPLC
- Complexity of samples (number of substances per sample)
  → processing time data evaluation
- Failure rate (technical faults and/or human errors)

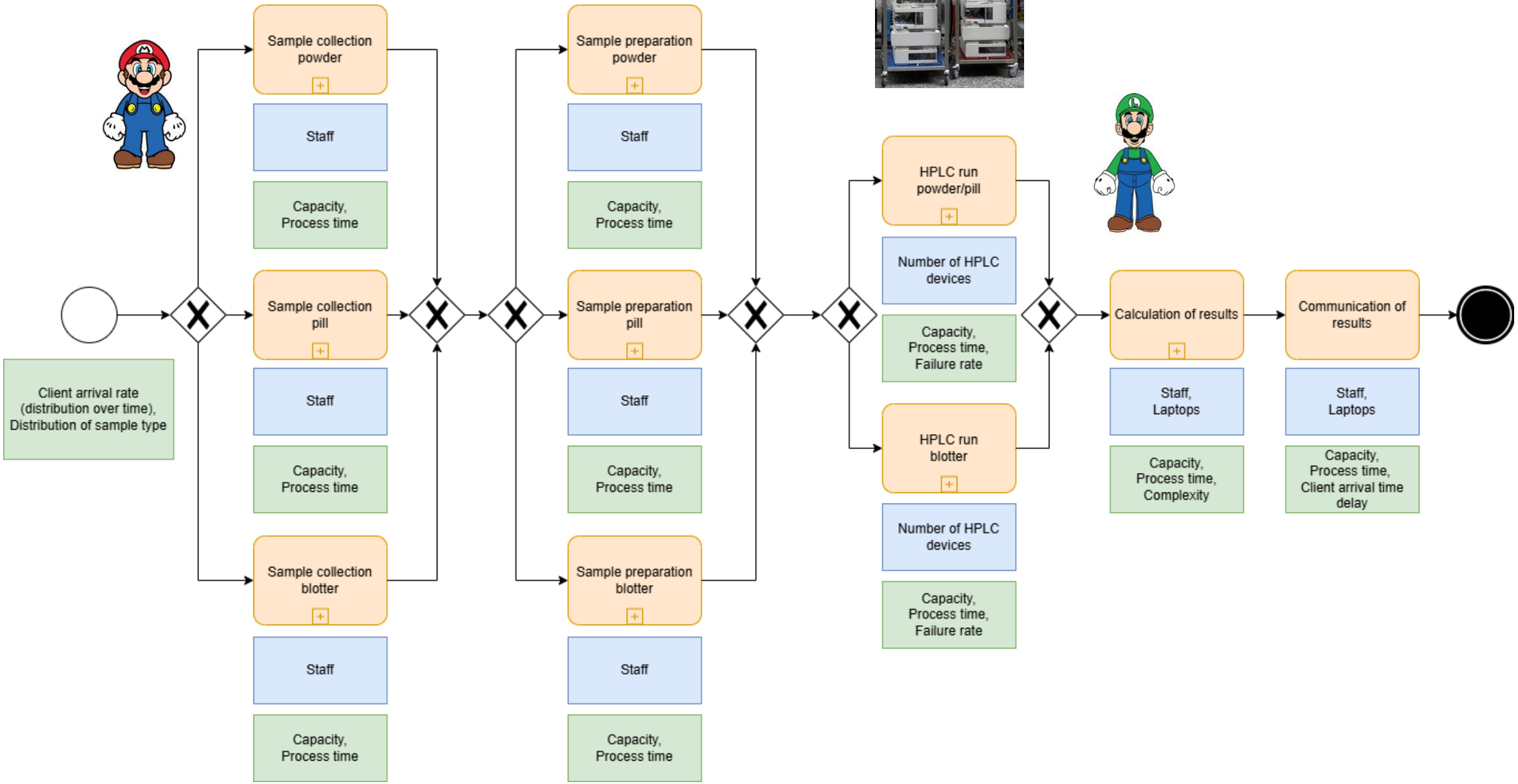
Stochastic elements

- Sample arrival rate per hour during the event
- Distribution / percentage of sample types
- Complexity of samples
- Failure rate (technical faults and/or human errors)

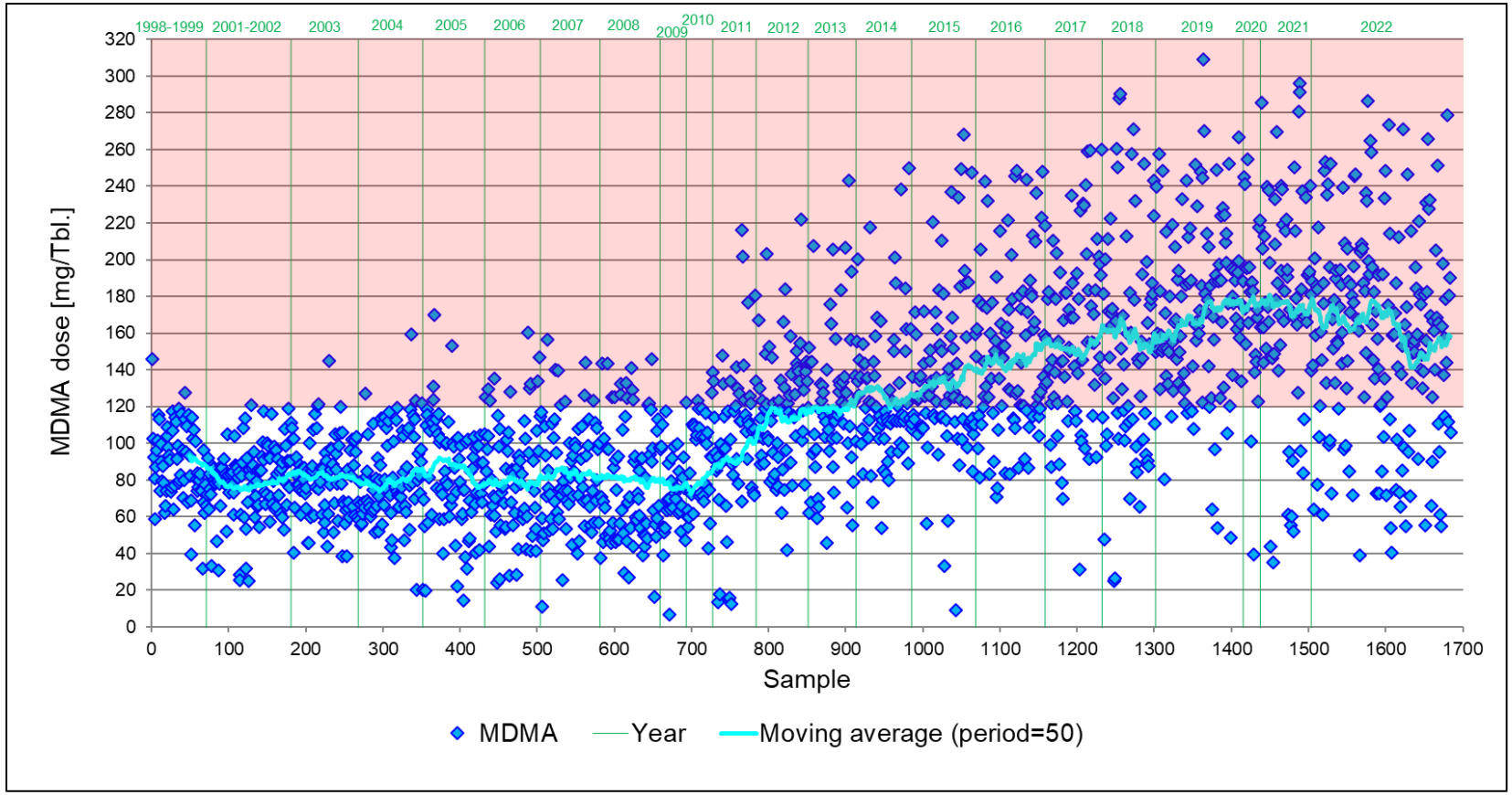
BPMN diagram without lanes



BPMN diagram + Parameters



Excurse: Variation of MDMA-dose in tablets



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